

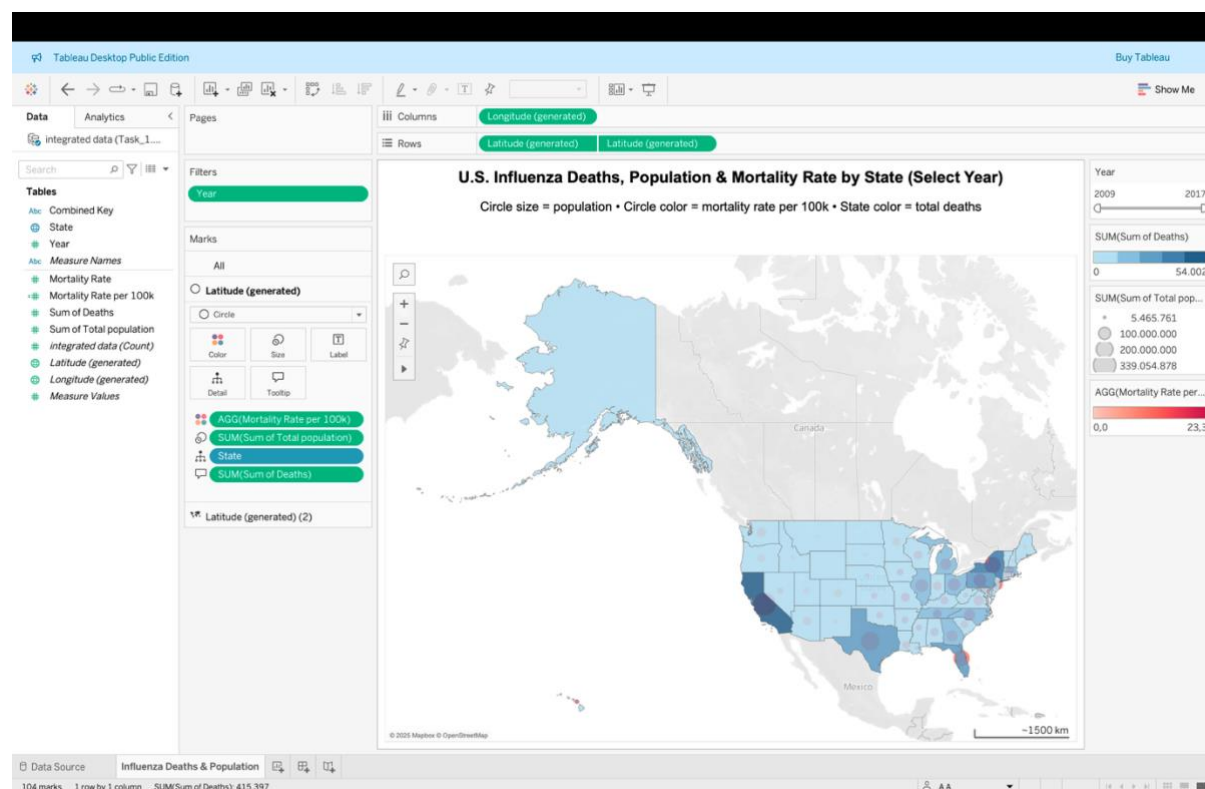
## Exercise 2.7 — Spatial Analysis (2009–2017)

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### Final Visualization (map summary)

**Title:** U.S. Influenza Deaths, Population & Mortality Rate by State (Select Year)

- State color (blue choropleth): total deaths (darker = more)
- Circle size: total population (larger = more people)
- Circle color (red scale): mortality rate per 100k (redder = higher per-capita risk)
- Year filter: 2009–2017
- Legends: deaths (color), population (size), mortality rate (color)



*Figure 1 Dual-axis state map (2009–2017): state color = total deaths; circle size = population; circle color = mortality rate per 100k; Year filter shown at right.*

## Written Answers

### 3) Spatial trends and hotspots (2009–2017)

- **Highest total deaths:** Large-population states are consistently darkest on the choropleth. California, Texas, and Florida stand out in most years. This aligns with their large populations (largest circles).
- **Lowest total deaths:** Smaller-population states (for example, Wyoming, Vermont, Alaska) remain light; totals are low, consistent with population size.
- **Per-capita hotspots:** Circle color highlights risk. Some smaller or mid-population states show redder circles in certain years, indicating higher mortality per 100k despite modest totals. This separates “big numbers” (population-driven) from “true risk” (rate-driven).
- **Time effects (Year filter 2009–2017):** The overall geography is stable across years; high-population states keep the highest totals. Redder circle pockets shift somewhat year to year, but no single year overturns the broad pattern. Stepping through the filter reveals local rate surges that do not always align with population and can inform staffing.

### Takeaway for staffing

Prioritize by **mortality rate per 100k** (circle color) rather than volume alone. States with redder circles—even if the state fill is only medium blue—represent higher per-capita risk and may merit targeted resources. States that are both dark blue and have red, large circles are the highest priority.

### Tableau Public Worksheet:

[https://public.tableau.com/views/Exercise2\\_7InfluenzaSpatialAnalysis20092017\\_AA/InfluenzaDeathsPopulation?:language=en-GB&publish=yes&:sid=&:redirect=auth&:display\\_count=n&:origin=viz\\_share\\_link](https://public.tableau.com/views/Exercise2_7InfluenzaSpatialAnalysis20092017_AA/InfluenzaDeathsPopulation?:language=en-GB&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link)